

Marketing Campaign Analysis

Objective

The objective of this task was to analyze marketing campaign data, understand customer purchasing behaviour, identify successful campaigns and recommend which marketing channel should receive more budget based on Return on Investment (ROI).

Data Analysis and Key Findings

The dataset contains 2,240 customer records. The analysis was performed using Python and Pandas after cleaning and preparing the data.

- 334 customers responded to the latest campaign, giving an overall response rate of 14.91%.
- Among the campaigns, Campaign 4 had the highest number of accepted responses (167), while Campaign 2 had the lowest (30).
- The Store channel recorded the highest number of purchases with 12,970, followed by Web with 9,150 and Catalog with 5,963.
- Wines had the highest customer spending among the product categories, with total spending of 680,816.
- Customers who responded to the campaign had higher average spending (987.39) compared with non-responders (538.93).

Marketing Channel and ROI Analysis

To compare the marketing channels, estimated marketing costs were introduced because the original dataset did not contain separate marketing costs for Web, Catalog and Store channels. The assumed costs used for the analysis were:

Web – 10,000 | Catalog – 12,000 | Store – 8,000

ROI was calculated using:

$$\text{ROI} = (\text{Revenue} - \text{Marketing Cost}) / \text{Marketing Cost} \times 100$$

The ROI comparison was used to identify the channel that could provide better returns for additional marketing investment. Based on the estimated ROI analysis, the Web channel showed strong return potential and is recommended for additional budget allocation.

Recommendation

Based on the analysis, I recommend increasing the marketing budget for the Web channel because of its strong estimated ROI and substantial purchase activity. The Store channel should also continue to be monitored because it generated the highest number of purchases. Future campaigns can focus more on customers with higher spending and on strategies used by the better-performing campaigns.

Conclusion

This analysis provided practical insights into customer behaviour, campaign performance, product spending and marketing-channel effectiveness. Using Python, Pandas and data visualizations helped convert the raw marketing data into useful business insights. The analysis can support better budget allocation and help improve the effectiveness of future marketing campaigns